

Fall 2025, 4100 Problem Set 3

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Given that $\mathbf{b}_1$ and $\mathbf{b}_2$ are two nonzero, linear independent vectors in $\mathbb{R}^3$ and that $T \in \mathcal{L}(\mathbb{R}^4, \mathbb{R}^3)$ is such that

$$T\mathbf{e}_1 = \mathbf{b}_1, \quad T\mathbf{e}_2 = -\mathbf{b}_1, \quad T\mathbf{e}_3 = 2\mathbf{b}_1, \quad \text{and} \quad T\mathbf{e}_4 = \mathbf{b}_2$$

find the general solution to $T\mathbf{x} = \mathbf{b}_1$

2. Let V be the vector space $\mathbb{P}_3(\mathbb{R})$ endowed with the inner product, $\langle x^i, x^j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$ and let S be a subspace of V given by, $S = \{f \in \mathbb{P}_3 : f(1) = f'(1) = 0\}$. Find bases for S and $S^\perp$.

Bonus: Find an integral formula for the above inner product

3. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{m \times l}$. Show that $\text{range}(A)$ is orthogonal to $\text{range}(B)$ if and only if $A^T B$ is the zero matrix.
4. Compute the fundamental subspaces of the following matrix

$$A = \begin{bmatrix} 1 & 0 & 4 & -3 & 2 \\ 4 & -2 & 1 & -12 & 3 \\ -3 & 2 & 3 & 9 & -1 \end{bmatrix}$$

5. Let

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

- a. Draw a directed graph having A as its incidence matrix
- b. Find bases for the four fundamental subspaces; $\text{range}(A)$, $\text{range}(A^T)$, $\text{null}(A)$, and $\text{null}(A^T)$
6. Let T be the linear map on $\mathbb{R}^2$ defined as follows

$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$
$$T \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

- a. Find $[T]_{\{\mathbf{e}_i\}}$ the matrix representation of T with respect to the standard basis.
- b. Let $\{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for $\mathbb{R}^2$ where $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$. Find $[T]_{\{\mathbf{v}_i\}}$, the matrix representation of T with respect to the basis $\{\mathbf{v}_i\}$ and use it write $T(4\mathbf{v}_1 + 5\mathbf{v}_2)$ as a linear combination of the basis vectors, $\mathbf{v}_i$.
7. Let $T : \mathbb{P}_2 \rightarrow \mathbb{R}_3$ be defined by $T(f) = f(-2)\mathbf{e}_1 + f(1)\mathbf{e}_2 + f(3)\mathbf{e}_3$. Find the matrix representation of T with respect to the standard bases of each space and use it to find the equation of the parabola passing through the points, $(-2, 8), (1, 3), (3, -5)$.